



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

May 19, 2025

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 21479
(FIRST REVISION)

EXPIRATION DATE: 2029-04-30

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Astra Space Operations, Inc.
Alameda, CA
2. PURPOSE AND LIMITATION:
 - a. This special permit authorizes the transportation in commerce of non-DOT specification aluminum lined composite cylinders incorporated into a propellant management system within spacecraft or components of spacecraft. The cylinders are based on the ISO 11119-2 Standard and are not equipped with pressure relief devices. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in the development of this special permit only considered the hazards and risks associated with the transportation in commerce.
 - c. No party status will be granted to this special permit.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR § 173.302a(a)(1) in that a non-DOT specification cylinder is authorized; and § 173.301(f)(1) and § 178.35(e) in that pressure relief devices are required, except as specified herein.
5. BASIS: This special permit is based on the application of Astra Space Operations, Inc. dated January 30, 2025, submitted in accordance with § 107.109.

Tracking Number: 2025024050

6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Xenon, compressed	2.2	UN2036	N/A

7. SAFETY CONTROL MEASURES: Prescribed packagings are non-DOT specification aluminum lined composite cylinders, without pressure relief devices, incorporated into a propellant management system and installed in spacecraft or components of spacecraft, which are overpacked in a transport container. The design of each cylinder holding xenon, is based on the ISO 11119-2 Standard. Details of the cylinders, components, and transport container are described in the November 18, 2022, application on file with the Office of Hazardous Materials Safety.

a. MARKING:

- (1) All cylinders must be plainly and durably marked “DOT-SP 21479”.
- (2) When cylinders are shipped installed in spacecraft or components of spacecraft, the outer transport container must be plainly and durably marked on two opposite sides in letters at least 2 inches in height on a contrasting background, “DOT-SP 21479” and “DO NOT STACK” as specified in § 172.301(c).

- b. OPERATIONAL CONTROLS: Cylinders may be shipped individually or integrated in a spacecraft or spacecraft components.

8. SPECIAL PROVISIONS:

- a. Under the terms of this special permit, the grantee may only offer hazardous materials (i.e., the grantee is not authorized as a carrier).
- b. A person who is not a holder of this special permit who receives a package covered by this special permit may reoffer it for transportation provided no modification or change is made to the package and it is reoffered for transportation in conformance with this special permit and the HMR.
- c. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.

9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, and cargo-only aircraft.
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel, aircraft, or motor vehicle used to transport packages covered by this special permit. The shipper must furnish a copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
 - o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, 49 CFR Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each “Hazmat employee”, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) —“The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. REPORTING REQUIREMENTS: Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this

special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:

A handwritten signature in blue ink, appearing to read "William Schoonover", is written over a faint circular stamp.

for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Material Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

PO: AS/MN